

The top down approach: The diagnosis starts from the operating system, device drivers, anti-virus software and application software, before going down towards the network access layer.

The bottom up approach: In this case, troubleshooting starts from the Ethernet media (cabling), Ethernet card, MAC address and IP address before moving up the layers towards the application layer. In short, this approach involves starting by locating faults in the network infrastructure before moving up to the application layer.

Before deciding upon the approach and starting the actual tests or hands-on diagnosis, discuss the problem with the end user or site engineer (if any). For example, when the problem of a slow network is reported, the actual problem could be non-accessibility (or slow speed) of a particular service (say, FTP). Here, you can certainly select the top down approach and proceed. As you troubleshoot more and more problems, you will develop 'troubleshooting insight', which helps to take the correct approach and locate the exact problem in the shortest possible time.

This article describes three real-life network troubleshooting scenarios. These will definitely be interesting, readily applicable and helpful during day-to-day network diagnostics, especially for field engineers.

Scenario 1: The Internet is not working

The network setup had approximately 75 computer systems spread across four floors in one building. The server room was located on the second floor, which housed the networking rack, the firewall, the router for Internet connectivity and the servers. Cabling was structured with patch panels in the rack and information outlets at the user end. Cables and passive network components were of a reputed brand. The network installation was approximately four years old. A total of six 10/100Mbps unmanaged Ethernet switches were used. Out of them, one switch connected all the servers and the remaining connected systems were located on various floors. Site inspection revealed that the cabling was done as per the industry's best practices—with numbering at both ends, cable bend radius guidelines being followed, and patch cable being used at both the ends.

Detailed discussions with the administrator revealed a very interesting problem. First, the Internet stops functioning, then the network goes on becoming slower and slower, and finally, it comes to a complete halt. Even local copying of files and accessing computer systems becomes impossible. The problem is random in nature and has developed gradually. Rebooting

the switches always stabilises the network, but after a random period of time, the problem reappears. To sum it up, the problem reported was actually the non-functioning of the entire LAN and was not limited to just "the Internet is not working."

The best approach to troubleshooting in this case would have been to use Wireshark if managed Ethernet switches with a SPAN port (port mirroring) were available. With the help of a SPAN port, traffic from all the other ports on that switch could be copied to one port. A system with Wireshark connected on this SPAN port would capture the entire switch traffic. An analysis of this would lead to the faulty component or system being detected. However, on this site, managed switches were not available; hence a different approach was followed:

1. First, the stacking of the switches was disconnected.
 2. Then, the Ping command was used continuously from one server to the default gateway.
 3. The stacking cables were connected, one by one.
- When one of the switches was stacked, the Ping response started slowing down—thus the faulty switch was isolated quickly. Proceeding further, to confirm the diagnosis, the faulty switch was replaced using an additional switch; but, unfortunately, the problem reappeared. So the fault was not with the switch. Troubleshooting continued further:
4. All Ethernet cables from the switch located in Step 3 were removed.
 5. One cable at a time was plugged – till the problem appeared.
 6. The computer system/device (which had a faulty Ethernet card!) with a slow Internet speed was isolated.

Scenario 2: Slow Internet speed

The set-up had 10 computer systems connected to the Internet using the GNU-based IPCop firewall via a DSL connection.

Discussions with the systems administrator revealed that the LAN speed was perfect; however, the Internet speed was very slow. Trials had already been conducted to bypass the firewall by connecting one laptop directly to the DSL connection. Here, the Internet speed available for one PC was satisfactory.

The discussions clearly revealed that the issue arose only when the office computer systems were trying to access the Internet. So, the following steps were taken to isolate the problem:

- a) The firewall was the first suspect – a firewall installed on a PC, and that too, with GNU? It must be the culprit! But bypassing it did not resolve the problem. Internet speed continued to be slow.
- b) Next, the possibility of a virus was explored.
- c) To capture complete network traffic, the Ethernet switch was replaced with an Ethernet hub (remember that

Table 1: TCP/IP layers, their functions and corresponding OSI layers

TCP/IP layers	Function	Corresponding OSI layer(s)
Application layer	Interacts with actual applications such as http, ftp, etc.	Application + presentation + session
Transport layer	Defines TCP/UDP protocols	Transport
Internet layer	Understands IP addresses	Network
Network access layer	Incorporates media and MAC addresses	Physical + data link